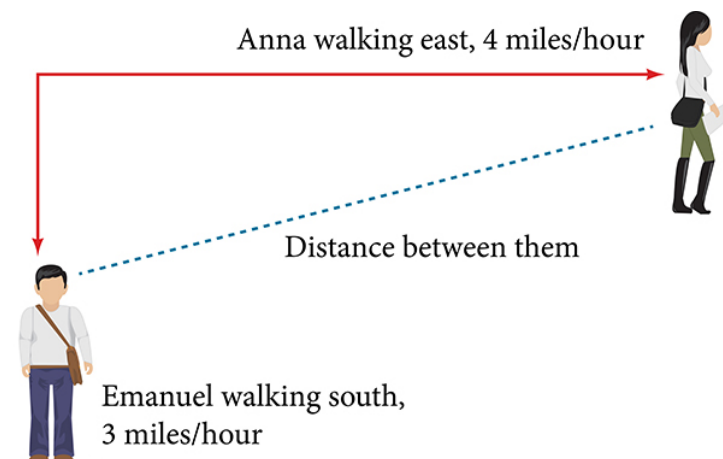


2.3: Modeling with Linear Functions

Example. Elan is a college student who plans to spend a summer in Seattle. Elan has saved \$3,500 for the trip and anticipates spending \$400 each week on rent, food, and activities. Write a linear model to represent this scenario. Identify and interpret:

- the rate of change
- the initial amount
- the x -intercept

Example. Anne and Emanuel start at the same intersection. Anna walks east at 4 miles per hour while Emanuel walks south at 3 miles per hour. They are communicating with a two-way radio that has a range of 2 miles. How long after they start walking will they fall out of radio contact?



Example. Jamal is choosing between two truck-rental companies. The first, *Keep on Trucking, Inc.*, charges an up-front fee of \$20, then 59 cents a mile. The second, *Move It Your Way*, charges an up-front fee of \$16, then 63 cents a mile. When will *Keep on Trucking, Inc.* be the better choice for Jamal?